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Leveraging Artificial Intelligence and Internet of Things for Enhanced Shuttle Tracking

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***Abstract*— In expansive workspaces and educational campuses, effective and dependable modes of transport are integral to smooth operations and mobility of people within campus. Shuttle buses serve as the only mode of commute for these sprawling school campuses linking different areas. Defects in shuttle operations such as long queues at the bus stop or uncertain arrival dates can be inconvenient and wasteful on users' side. This project aims at making campus shuttles more efficient and user-friendly through real-time tracking, predictive arrival times, and operational analysis. By using the Global Positioning System (GPS) technology along with Internet of Things (IoT) devices, the system display the live shuttle locations as well as predict arrival times while also analyzing the shuttle speed with the aim of optimizing routes alongside fuel consumption. The objective is therefore reducing waiting time, improving commuting experience in general while contributing towards sustainability in transportation systems within campuses.**

Keywords*— *Shuttle buses, ESP32, Internet of Things, Real-time tracking, Campus commute, Route optimization.

I. Introduction

Extensive campuses, be it within academic circles or business environments, depend greatly on internal transportation systems to guarantee effective movement for students, employees, and guests. Shuttle buses, which are frequently used and economical, can encounter issues like lengthy waiting periods, absence of live updates, and inconsistent timetables, resulting in dissatisfaction and time wastage for users.

This project tackles these problems by incorporating cutting- edge technologies such as GPS tracking and IoT analytics into campus shuttle systems. The system enables users to better plan their commutes by integrating GPS modules on shuttles, providing live location updates through a mobile app or web platform. Using past and current data, predictive algorithms predict shuttle arrival times to decrease uncertainty. Moreover, IoT devices track shuttle velocity and fuel usage, transmitting information to cloud-based analytics platforms for improved route planning and operational effectiveness.Our goal is to develop a dependable shuttle service for large workplaces and schools.

The positive aspects of the newly proposed method consist of:

- Reduced Delays.
- Enhanced User Experience.
- Optimized shuttle routes and schedules to improve operational efficiency.
- Environmental sustainability.
- Cost Efficiency.

II. Related Works

D. Sharma et al.[1] posed an embedded system which is used for tracking and positioning of any vehicle by using GPS and GSM. In this solution 8052 microcontroller is used for interfacing to various hardware peripherals.The microcontroller is interfaced serially to a GSM Modem and GPS Receiver. A GSM modem is used to send the position of the vehicle from a remote place. The GPS modem will continuously give the data indicating the position of the vehicle. At the same time, the use of EEPROM for data storage limits the amount of data that can be stored. This would fail to store the tracking data over a long period. The 8052 microcontroller has limited processing power and memory, which may restrict the ability to add more complex features or handle large amounts of data efficiently.The use of multiple components like the GPS modem, GSM modem, MUX, and EEPROM increases the system's hardware complexity that indeed increases the overall maintenance cost.

Kushal Gogri et al.[7] shaped an android application that gives necessary information about all the buses travelling in Mumbai.They designed a tracking system that will help the user to check out the current location of the bus and also will help the user to know how much

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time the bus will take to reach the current location of the user.The proposed framework is specifically developed for various metropolitan cities such as New Delhi, Mumbai, which restricts its usability in rural areas where arrival information might be needed.The system needs reliable internet connectivity,areas having poor internet connectivity can delay some important updates for the user.

C.K. Gomathy et al.[9] presented a method where Global Po- sitioning System [GPS] is used for a Solar Assisted Rickshaw Van using Arduino UNO R3 and GSM module information about the location of the vehicle that is to be monitored and then send the latitude and longitude to the monitoring center through satellite.At the monitoring center different software is used to display the vehicle on the Google map.Monitoring center software helps the vehicle owner with a view of the vehicle location on an electronic map.Despite its advantages, this system still faces a number of limitations, such as the use of SMS communication in the proposed methodology can delay various potential issues with message delivery.Network delay can yield delays in sending and receiving position updates of the van.The Google Map API usage involves cost,especially when the system makes a large number of requests.This could increase the overall cost of the system.The Arduino Uno R3 also has limited processing power and memory, which may restrict the complexity and scalability of the system.

M.H. Zohari et al.[10] introduced a novel approach that utilizes Arduino MEGA as a microcontroller and Ublox NEO-6M GPS module which are used to route the coordinates while SIM900A GSM module is used to connect with the user. This system can position the location of a vehicle and show it on Google Maps. In general, this system will send the coordinates and Google Maps link to mobile phone users using messages.The proposed solution fails to obtain accurate coordinates indoors or in areas with obstructed views of the sky, as the GPS module struggles to maintain a direct signal connection with the satellite. The Ublox NEO-6m GPS module fails to provide the highest level of accuracy compared to more advanced GPS modules.

Jimmy Wong Wei Min et al.[12] proposed a system developed using a web-based and an IoT platform (Arduino), which acts as a tracking device.This system is developed for all vehicle users to provide easy tracking of their vehicle location.The GSM module in this framework is responsible for sending the location to the users via Short Message Service (SMS) or data transfer to the web-server.However, the use of SMS communication can delay various potential issues with message delivery and further network delay can yield delays in sending and receiving the coordinates.Also, the need for users to interact with a web-based application adds a layer of complexity. Users operating these web services from rural areas might find this challenging.

Kute Sushma Vijay et al.[14] proposed a GPS-based bus tracking method which sends the coordinates of the bus either by a GPRS service or a RFID directly at the bus stops using RF receivers.The proposed system is further integrated with the historical average speeds of each segment. This is done to improve the accuracy by including the factors like volume of traffic, crossings in each segment, day and time of day.However,the bus tracking which is monitored using various factors like volume of traffic that can altered by numerous unprecedented incidents such as road accidents which may not be reflected in the historical averages and may yield to data inconsistency.This system also fails to provide data accuracy as the RFID systems have a limited range of operation of about 100 m.

III. Proposed Methodology

A. Hardware and Software

1) GPS Module: The GPS module is responsible for obtaining the real-time location (latitude and longitude) of the bus. This module receives signals from GPS satellites and calculates the exact location of the bus. It communicates with a microcontroller (ESP32 in this case) to ensure the bus position is constantly updated. This model can provide information such as speed, altitude, and number of satellites in view, which can be used to improve the accuracy of location tracking.

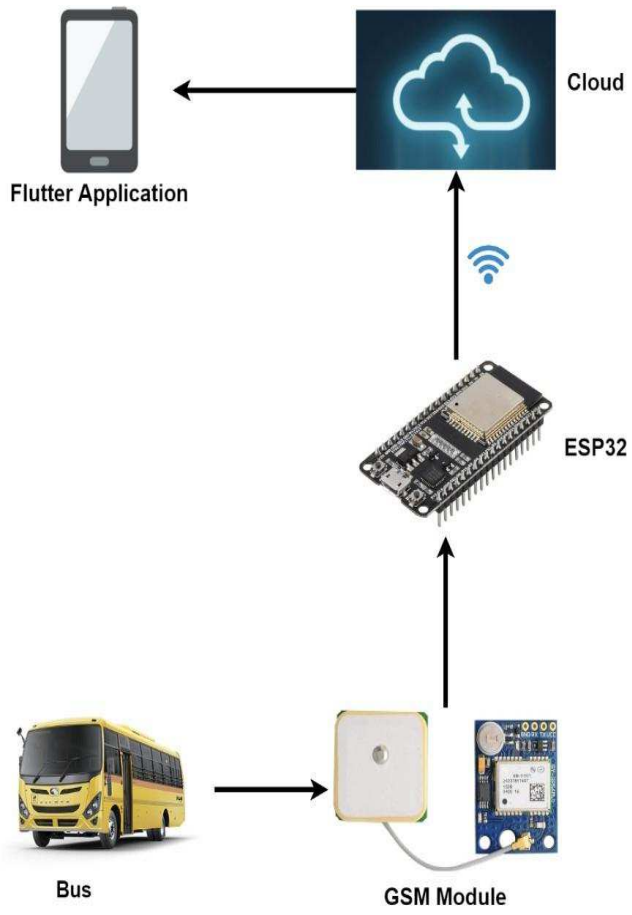


Fig. 1. System Architecture

1) ESP32: ESP32 microcontroller works as a central controller on the bus. It receives location information from the GPS module and manages the transmission of this information to the cloud via Wi-Fi. It is frequently used in IoT applications due to its versatility, low power consumption and ability to perform multiple tasks at the same time. In this system, ESP32 processes the data it receives from the GPS module and sends it to the cloud using the MQTT protocol or HTTP, ensuring that the data is sent quickly and efficiently.

2) Wi-Fi: Wi-Fi plays a key role in the design, allowing the ESP32 microcontroller to send real-time GPS data to a cloud server. The ESP32's built-in Wi-Fi module connects to the local network to ensure continuous data transmission and efficient operation. This connection allows the shuttle's location to be updated in real time to the cloud, which passengers can access via a

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mobile app. As a means of transportation, secure Wi-Fi coverage is critical to maintaining uninterrupted communications and providing users with accurate, up-to-date information.

3) Cloud: The cloud acts as a central storage and processing center for all data sent by ESP32. It stores location data, functions and makes it accessible via mobile devices. The cloud infrastructure is designed to scale according to the amount of data processed to ensure that the system continues to operate even with multiple data paths and multiple users. The cloud can also facilitate data analysis by running algorithms on stored data, such as developing bus routes and predicting bus arrival times.

Flutter app: Flutter mobile app is a user interface for passengers to get real bus information. The app allows users to view bus locations, estimated arrival times, and other relevant information on their devices. The app communicates with the cloud via RESTful API to retrieve data and display it intuitively and efficiently. The app can also provide other features such as route planning, notifications, and feedback.

B. System Architecture

This prototype design demonstrates an instantaneous bus solution using IoT technology. The setup starts with a GPS module installed on the bus that continues to capture the exact location of the vehicle. This data is then sent to the ESP32 microcontroller, a powerful device that can manage Wi-Fi communication and process the data. ESP32 acts as an intermediary by sending the location data to the cloud server for central storage and further processing. Leveraging Wi-Fi connectivity, the system allows the bus location to be updated in real time. This cloud-based system makes the system scalable and accessible from anywhere with internet access. The app retrieves location data from the cloud and displays it in a user-friendly manner, allowing users to track bus movements and predict arrival times. This increases the convenience and efficiency of transportation, especially for students and passengers who rely on bus schedules. The system offers solutions to today's transportation challenges by integrating IoT devices with cloud computing and mobile applications.

A. C. Entity Association Model (EAM)

The Entity Association Model (EAM), sometimes referred to as an EAM, is a graphical representation that illustrates the connections between entity sets within a data repository. It models actual-world objects and their associations. This diagram uses specific symbols and notations to represent these relationships, including:

- Boxes representing entity groups.
- Diamonds depicting relationship groups.
- Lines connecting attributes to entity groups, as well as linking entity groups to relationship groups.
- Attributes designated as primary keys are identified by under- lining.

The enhanced shuttle tracking system's ER diagram captures the key entities and their relationships, illustrating how the system operates. The Shuttle entity, which represents the vehicles in the system, interacts with the Route entity, defining the paths the shuttles take, and the Schedule entity, which manages the timing of these operations. Shuttles are equipped with devices, represented by the Device entity, that monitor parameters like speed and fuel consumption. The User entity represents individuals using the shuttle service, and their interaction with the system is recorded through the User-Schedule entity, linking users to specific schedules.

Additionally, the system includes Prediction and Analytics entities.Prediction generates expected arrival times based on the schedule, while Analytics aggregates data on shuttle performance, such as average speed and fuel efficiency. The relationships between these entities are crucial, with some, like User-Schedule and Prediction, being weak entities dependent on their associated entities for existence. This interconnected structure ensures a comprehensive and efficient management of the shuttle operations, user interactions, and system performance.

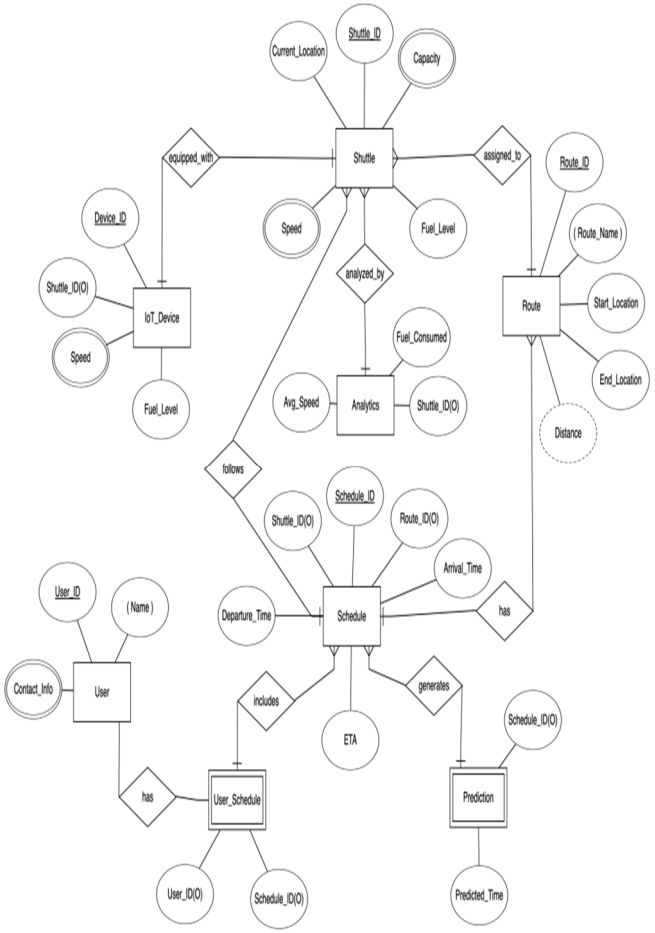


Fig. 2. ER Diagram

IV. System Developments and Results

A. A. Circuit Diagram

The NEO-6M GPS module is a highly reliable GPS receiver that is compatible with a wide range of microcontroller boards, making it an ideal choice for projects requiring precise location tracking. The module is capable of receiving data related to location coordinates, speed, altitude, and time, which are crucial parameters for applications in navigation, tracking systems, and mobile robotics.

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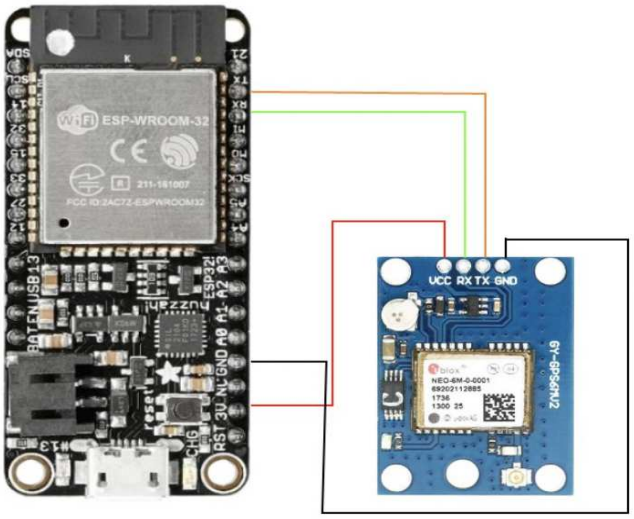


Fig. 3. Circuit Setup

The NEO-6M GPS module is connected to an ESP32 microcontroller,allowing it to receive GPS data and process it.

VCC to 3V3 (Red Line): The VCC pin of the GPS module is connected to the 3.3V pin on the ESP32. This powers the GPS module.

GND to GND (Black Line): The GND pin on the GPS module is connected to the GND pin on the ESP32. This completes the circuit and provides a common ground reference.

TX to RX2 (Green Line): The TX pin on the GPS module is connected to the RX2 pin on the ESP32.

RX to TX2 (Orange Line): The RX pin on the GPS module is connected to the TX2 pin on the ESP32. This connection is typically used for sending data from the ESP32 to the GPS module, though in most GPS applications, this is not required.

A. *Mobile Application*

1) Driver Application: The User Interface(UI) shown in the image is designed for a shuttle bus driver, offering essential in- formation for route management and navigation. At the top, there is a status bar displaying the time and battery level. Below it, a prominent label indicates the ”Next Stop in 05:00,” alerting the driver of the time remaining until the next scheduled stop.

The central part of the screen features a map, highlighting the bus’s current location, the upcoming destination, and surrounding landmarks, including the SRM University campus and the SRM lake. A bold green circular button labeled ”GO” is positioned at the center of the map, likely functioning as a start or confirmation button for the driver’s next action or journey. At the bottom of the screen, the UI presents a summary of key metrics: ”Main Campus Visits” with a count of 12, ”Annexe Campus Visits” with 11, and ”Stops before Next Refuel” with 3. This section helps the driver track their progress on the route and manage fuel.

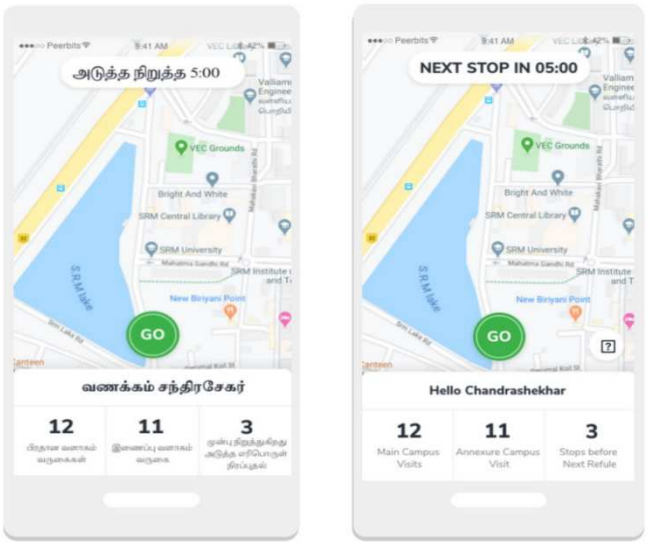
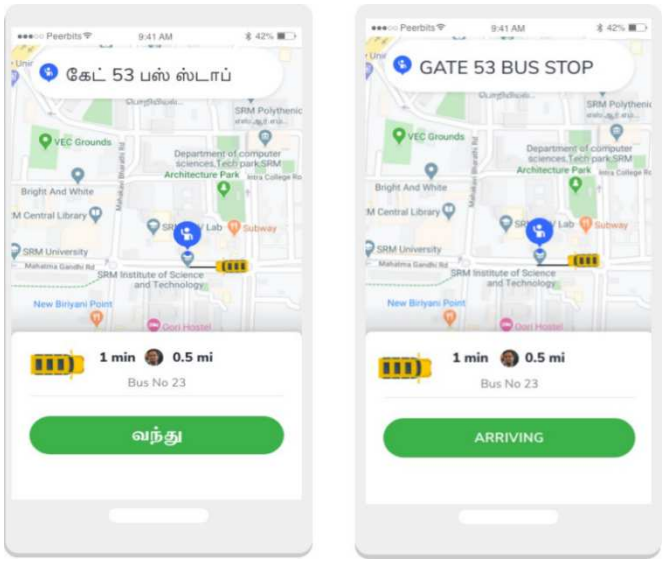


Fig. 4. Driver Application

2) User Application: The user interface (UI) displayed is part of a mobile application designed for tracking the arrival of shuttle buses, particularly for college students. The central feature of this UI is the map, which visually presents the user’s location in relation to nearby landmarks and shuttle routes. Prominently marked on the map is ”GATE 53 BUS STOP,” indicating the specific location where the user is expecting the bus. The UI also provides real-time information about the shuttle’s proximity, displaying that the next bus, identified as ”Bus No 23,” is just 1 minute away



and 0.5 miles from the stop.

Fig. 5. User Application

At the bottom of the screen, a green button labeled ”ARRIVING” highlights the immediacy of the bus’s approach, signaling to the user that they should prepare to board. This clear, concise presentation of information, including the map view, bus details, and the arrival notification, ensures that users can easily determine when the shuttle will arrive, enhancing their overall commuting experience. The UI is designed to be user-friendly, providing

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essential details at a glance, which is critical for students who rely on timely and accurate bus schedules.

C. Results

The interface for shuttle drivers offers live information on the vehicle's position and the remaining distance to the upcoming stop or end point.This interface is designed to ensure the driver is aware of their current position and can adjust routes or schedules accordingly.As demonstrated in the image captured from the Serial Monitor of the microcontroller, the system successfully processes and transmits NMEA sentences.



Fig. 6. Arduino IDE

These sentences contain vital information such as latitude, longitude, ground speed, course over ground, and time of fix. For example, the GPRMC sentence contains data on the current coordinates (latitude 41.0732627 and longitude -73.79323), indicating the shuttle's precise location. The information also includes speed and time, which are crucial for calculating the distance to the next stop and updating this data on the user interface in real-time.

V. Conclusion

The project successfully improves the shuttle bus system leveraged in most of the large institutions and work places by integrating GPS tracking and IoT analytics, solving issues such as long wait times, lack of timely updates, and inconsistent schedules. Users can plan their journeys more efficiently, reducing uncertainty and delays by providing real-time location updates and estimated arrival times via mobile devices. This helps improve operations, services, and sustainability. Overall, the program provides a reliable, efficient, and environmentally friendly shuttle service that benefits customers and the environment.

VI. Future Work

Shuttle tracking systems present numerous possibilities for further research and innovation. Several prospective areas for future explo- ration are:

1) Coordination with Various Transportation Methods: Extend the system to include tracking for multiple types of vehicles, such as trams, trains, or even ride-sharing services,providing a unified transportation app.

2) Real-Time Traffic Analysis: Incorporate real-time traffic data to dynamically adjust bus routes or provide alternative route suggestions to minimize delays.

3) Passenger Load Monitoring: Add sensors to monitor passenger load in real-time, allowing users to see how full a bus is before it arrives and enabling better crowd management.

4) Custom Alerts and Notifications: Develop personalized notifi- cations within the app, such as alerts for when a bus is nearing a user's location or when a service disruption occurs.

5) Security Enhancements: Strengthen data encryption and imple- ment advanced authentication methods to safeguard the confidentiality and protection of user data and location details.

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